

Abstract algebra – 5

Representations and the averaging operator

Homework

13 сентября 2026 года

Theoretical minimum. A representation of a finite group G is a homomorphism $\rho: G \rightarrow \mathrm{GL}(V)$ over $\mathbb{C}$; its character is $\chi(g) = \mathrm{tr} \rho(g)$. The operator $P = \frac{1}{|G|} \sum_g \rho(g)$ satisfies $\rho(h)P = P$, hence $P^2 = P$, its image is the space V^G of invariants and $\dim V^G = \frac{1}{|G|} \sum_g \chi(g)$; if there are no invariants then $\sum_g \rho(g) = 0$. Averaging an inner product shows that every finite subgroup of $\mathrm{GL}_n(\mathbb{C})$ is conjugate to a unitary one, so all its elements are diagonalisable with eigenvalues roots of unity; averaging a projection proves Maschke's theorem, so every representation splits into irreducibles. Schur's lemma says that an intertwiner of irreducibles is zero or invertible, and is scalar when the source and target coincide. Applying the trace formula to $\mathrm{Hom}(V, W)$ gives $\dim \mathrm{Hom}_G(V, W) = \langle \chi_W, \chi_V \rangle$, so irreducible characters are orthonormal; the regular representation gives $\sum_i (\dim V_i)^2 = |G|$, and the number of irreducible representations equals the number of conjugacy classes.

1. Let $n > 4$ be even and let $G \leq \mathrm{GL}_2(\mathbb{C})$ be a subgroup with $|G| = n$. Prove that there exists H with $\{I_2\} \neq H \neq G$ such that $XYX^{-1} \in H$ for all $X \in G$ and all $Y \in H$. (Румыния, 2019)
2. Compute the character of the regular representation and deduce both $\sum_i (\dim V_i)^2 = |G|$ and $\sum_i (\dim V_i) \chi_i(g) = 0$ for $g \neq e$. Deduce also that $\dim V_i$ divides $|G|$ in the two smallest non-abelian cases S_3 and Q_8 , and check the identity on them.
3. Prove that all irreducible representations of a finite abelian group are one-dimensional, and that a group possessing an irreducible representation of dimension d has order at least d^2 . Then prove that the number of irreducible representations equals the number of conjugacy classes.

4. Show that the averaging trick genuinely needs finiteness: the group generated by $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ in $\mathrm{GL}_2(\mathbb{R})$ is not conjugate to any group of unitary matrices. Identify precisely which step of the proof of Theorem 2 fails, and give an example of an infinite subgroup of $\mathrm{GL}_2(\mathbb{C})$ for which the conclusion nevertheless holds.
5. Prove that if a finite group G has a faithful irreducible representation, then its centre $Z(G)$ is cyclic, and give an example showing that the converse fails. (The related IMC problem on anticommuting idempotent matrices is Linear algebra – 2, problem 8.)
6. Let ρ be a representation of a finite group G with character χ . Prove that if $\chi(g) = \chi(e)$ for some g , then $\rho(g)$ is the identity, and that if $|\chi(g)| = \chi(e)$, then $\rho(g)$ is a scalar matrix. Deduce that $\{g : \chi(g) = \chi(e)\}$ is a normal subgroup.
7. Let G act on a finite set X , and let V_d be the space of G -invariant homogeneous quadratic forms on $\mathbb{C}^X$. Prove that $\dim V_2 = \frac{1}{2|G|} \sum_g (\chi(g)^2 + \chi(g^2))$, and compute this dimension for the rotation group of the cube acting on the six faces.
8. For conjugacy classes C_1, C_2 of a finite group and $g \in G$, prove that the number of ways to write $g = xy$ with $x \in C_1, y \in C_2$ equals

$$\frac{|C_1||C_2|}{|G|} \sum_{\chi} \frac{\chi(C_1)\chi(C_2)\overline{\chi(g)}}{\chi(e)},$$

the sum being over the irreducible characters.

9. Prove that a finite group G acting on a finite set X transitively, with $|X| > 1$, has a representation $\mathbb{C}^X = \mathbf{1} \oplus W$ with $\dim W = |X| - 1$, and that W is irreducible exactly when the stabiliser of a point has two orbits on X . Apply this to S_n acting on $\{1, \dots, n\}$.
10. Let $G \leq \mathrm{GL}_n(\mathbb{C})$ be finite and let $\chi(g) = \mathrm{tr} g$. Prove that $\frac{1}{|G|} \sum_g |\chi(g)|^2$ is a positive integer, and that it equals 1 if and only if the tautological

representation of G is irreducible. Use this to decide whether the rotation group of the cube acts irreducibly on $\mathbb{R}^3$.